ity assurance by facilitating remote audits and process improvement. Automobile heavyweights are using metaverse precursor technologies to create and test designs collaboratively," says Immanuel J. Kingsley, Metaverse Technology Officer, and Head of Innovation Lab at Hexaware Technologies.

Sadaf Siddiqui, General Manager - Industry Marketing, Keysight Technologies says, "We are talking about a massive industrial revolution which will be enabled by 5G. Some of the exciting pointers are predictive maintenance—to reduce downtime, and efficient analytics—to optimise resources and improve performance. The metaverse aspect will bring in remote connectivity option as well, wherein multinational setups can then be called as ideal global setups."

Garg asserts that the industrial metaverse is relevant to every subsegment of industrial companies across a range of use cases. "For example, manufacturers of complex and large industrial machines can demonstrate their product in a virtual factory environment within the metaverse, without having to worry about transporting this equipment to different demonstration locations. To enhance productivity and safety, mining companies or equipment manufacturers for the mining industry can create a digital twin of their operations or of their products. Similarly, manufacturers of industrial products, whether fasteners or chemicals, can virtually train workers who handle their products in correct application and usage while creating what-if simulations to teach the handling of special scenarios," he says.

Tech components of the metaverse

"Digital twin is strategic as it helps to understand how the physical counterpart will behave under diverse set of operating conditions. Some of the other key components can be AI, VR,

Deutsche Bahn's mixed reality solution to efficiently and safely train engineers on railway parts and maintenance (Courtesy: Microsoft)

software intelligence, data analytics. There is no pre-defined standard model—it depends on what level of intelligence or automation we want to introduce into the segment. This can also be scaled up or down depending on the requirements and needs," says Siddiqui, adding that the crux remains around smart connectivity with a high level of software intelligence.

Garg explains that, "The metaverse is a virtual collective shared space that is persistent, interactive, and immersive. It can be public or private, depending on the access privileges of its users. Technologies such as extended reality, blockchain, the IoT, and AI form the core of a metaverse ecosystem." Extended reality (XR) is an umbrella term that covers virtual reality (VR), augmented reality (AR), and mixed reality (MR).

Digital twins. Very simply put, a digital twin is a virtual representation of a physical entity—product or process. Sensors capture operational and environmental data and stream them to the digital twin on the cloud. This is augmented with enterprise systems data—manufacturing execution systems (MES), enterprise resource planning (ERP), computer-aided design (CAD) models, etc. The physical and virtual realms meet using edge computing technologies. Artificial intelligence (AI) discovers insights from the data—the more data there is, the better the AI system becomes. Feedback goes back to the physical world through actuators and decoders. This is the operation of a simple digital twin. In the industrial metaverse, multiple digital twins will be integrated.

By using digital twin technology right from the design phase, it is possible to test and validate designs of products and processes to mitigate errors. Boeing is building its next major plane using digital twin tech. In a press release, the company explained that 70% errors arise in the design stage. So, if more and more errors and discovered—and rectified—in the digital twin, then there will be less possibility of errors in the final product.

Current technologies, like that made possible by the Siemens Nvidia collaboration, enable photo-realistic, physics based representations of the real world. For example, if you simulate a difference in temperature or a part failure, the digital twin will react exactly like the physical asset would.

McKinsey predicts that digital twins will be a $48 billion business by 2026. A Grand View Research report suggests that the digital twin market may reach $86.09 billion by 2028, fuelled by Industry 4.0 use cases.

The Internet of Things. The industrial metaverse is nothing without the IoT—a fast, interconnected system of edge devices that collects and distributes data via the cloud. According to a report by IoT Analytics, the global number of connected IoT devices was 14.4 billion as of May 2022, and likely to go up to more than 27 billion connections by 2025. A lot of research is happening on reducing the size, power consump-